

*Proof.* **Degenerate and edge cases.**

*Case  $n = 1$ .* The hypothesis gives  $\lambda_1 = 1$ ; hence the statement reads  $f(x_1) \leq f(x_1)$ , which holds with equality.

*Zero weights.* If  $\lambda_k = 0$  for some index  $k$ , then the term  $\lambda_k x_k$  (resp.  $\lambda_k f(x_k)$ ) disappears from the left-hand (resp. right-hand) side. Thus the inequality for  $\{x_i, \lambda_i\}_{i=1}^n$  is equivalent to the inequality for the reduced family obtained by omitting  $k$ . Hence we may assume  $\lambda_i > 0$  for all  $i$ .

**The case  $n = 2$ .**

Let  $\lambda_1, \lambda_2 \geq 0$  with  $\lambda_1 + \lambda_2 = 1$ . By convexity,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \quad (t \in [0, 1], x, y \in \mathbb{R}).$$

Choosing  $t = \lambda_1$ ,  $x = x_1$ ,  $y = x_2$  yields

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2),$$

which is Jensen's inequality for two points.

**Inductive hypothesis.**

Assume that for some integer  $k \geq 2$  and for every choice of points  $y_1, \dots, y_k$  and weights  $\mu_1, \dots, \mu_k$  with  $\mu_i \geq 0$  and  $\sum_{i=1}^k \mu_i = 1$  we have

$$f\left(\sum_{i=1}^k \mu_i y_i\right) \leq \sum_{i=1}^k \mu_i f(y_i). \quad (\text{IH})$$

**Inductive step  $k \rightarrow k+1$ .**

Let  $\lambda_1, \dots, \lambda_{k+1} \geq 0$  satisfy  $\sum_{i=1}^{k+1} \lambda_i = 1$ . Set

$$\alpha := \lambda_{k+1}, \quad \beta := 1 - \alpha = \sum_{i=1}^k \lambda_i.$$

If  $\beta = 0$  then  $\alpha = 1$  and the desired inequality reduces to  $f(x_{k+1}) \leq f(x_{k+1})$ . Hence we may assume  $\beta > 0$ .

Define the intermediate point

$$y := \frac{1}{\beta} \sum_{i=1}^k \lambda_i x_i.$$

The numbers  $\frac{\lambda_i}{\beta}$  ( $i = 1, \dots, k$ ) are non-negative and sum to 1, so  $y$  is a convex combination of the first  $k$  points.

*4.1 Applying two-point convexity.* Since  $\alpha \in [0, 1]$  and  $\beta = 1 - \alpha$ , convexity gives

$$f(\beta y + \alpha x_{k+1}) \leq \beta f(y) + \alpha f(x_{k+1}). \quad (1)$$

Moreover,

$$\begin{aligned}\beta y + \alpha x_{k+1} &= \beta \left( \frac{1}{\beta} \sum_{i=1}^k \lambda_i x_i \right) + \lambda_{k+1} x_{k+1} \\ &= \sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1} = \sum_{i=1}^{k+1} \lambda_i x_i.\end{aligned}\tag{2}$$

Thus the left-hand side of (1) is precisely the term we need to bound.

*4.2 Estimating  $\beta f(y)$  via the induction hypothesis.* Applying (IH) to the points  $x_1, \dots, x_k$  with weights  $\frac{\lambda_i}{\beta}$  yields

$$f(y) = f\left(\sum_{i=1}^k \frac{\lambda_i}{\beta} x_i\right) \leq \sum_{i=1}^k \frac{\lambda_i}{\beta} f(x_i).\tag{3}$$

Multiplying (3) by  $\beta > 0$  gives

$$\beta f(y) \leq \sum_{i=1}^k \lambda_i f(x_i).\tag{4}$$

*4.3 Combining the estimates.* From (1) and (2) we have

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \beta f(y) + \alpha f(x_{k+1}).$$

Using (4) and  $\alpha = \lambda_{k+1}$  we obtain

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i),$$

which completes the inductive step.

**Conclusion of the induction.** The inequality holds for  $n = 2$  (Section2) and the step  $k \rightarrow k + 1$  has been proved; therefore, by induction, Jensen's inequality holds for every integer  $n \geq 2$ . Together with the trivial case  $n = 1$  (Section1) we obtain the result for all  $n \in \mathbb{N}$ .

**Equality conditions.**

1. If one weight equals 1 and the others are zero, both sides reduce to  $f(x_j)$  for that index  $j$ .
2. If  $f$  is affine on the convex hull of the points  $\{x_i \mid \lambda_i > 0\}$ , i.e. there exist  $a, b \in \mathbb{R}$  such that  $f(t) = at + b$  on that hull, then equality holds.
3. In all other situations at least one of the intermediate convexity inequalities used in the inductive step is strict, so the overall inequality is strict.

□